# GenoFold: A Confidence-Aware, Explainable, Multi-Modal Framework for 3D Genome Architecture Reconstruction

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***Abstract*—Chromatin folding sets which enhancers reach their target promoters and where the boundaries of topologically associating domains fall, so reconstructing genome architecture in three dimensions from Hi-C has become a standard step in reading regulatory structure. Existing reconstructors share three limitations: they use Hi-C alone, they return coordinates with no calibrated estimate of reliability, and their preprocessing, inference, and scoring stages live in separate tools. GenoFold addresses all three within one trained pipeline. It encodes genomic bins as a contact graph that carries Hi-C alongside sequencing coverage, sequence composition, and mappability; pairs a four-layer GATv2 encoder with a six-layer Transformer for local and long-range context; and decodes 3D coordinates under a self-supervised distance-matching objective. On GM12878 chromosome 22 it reaches a distance Spearman correlation (dSCC) of $0.893 \pm 0.001$, level with HiC-GNN and within a hundredth of 3DMax. GenoFold matches established methods on dense data, while its key advantage lies under sparse Hi-C, where multi-modal features hold up as the contact graph thins. Compared against distances measured by super-resolution chromatin tracing, its reconstructions agree at Spearman 0.986 (IMR90) and 0.981 (K562). The methods separate under shallow Hi-C: with contacts downsampled to 5%, GenoFold still recovers the measured structure at 0.978/0.969, while 3DMax falls to 0.819/0.831. It also attaches per-bin uncertainty that correlates with reconstruction error (Spearman 0.28) and transfers to unseen cell lines without retraining.**

***Index Terms*—3D genome reconstruction, Hi-C contact maps, graph attention networks, multi-modal genomic learning, uncertainty quantification, explainable AI**

## I. Introduction

The spatial folding of chromatin is inseparable from genome function. It determines which enhancers can physically reach their target promoters, sets the boundaries of topologically associating domains (TADs), and separates the nucleus into transcriptionally active and inactive compartments [1]. When this organization is disturbed, the consequences are concrete: mislocalized regulatory contacts have been tied to developmental disorders, oncogenesis, and dysregulation of immune-response genes [2].

The primary experimental signal for reconstruction is Hi-C, a proximity-ligation assay that measures contact frequency between pairs of genomic loci genome-wide [3]. Existing methods have progressed along several lines: 3DMax treats contact frequency as a proxy for Euclidean distance and optimizes coordinates by maximum likelihood [4]; HiC-GNN applies graph neural networks to the contact matrix, learning structural representations that transfer across resolutions [5]; FLAMINGO addresses sparsity through low-rank completion and multi-scale inference [6]; ShRec3D frames reconstruction as shortest-path embedding followed by classical multidimensional scaling [7]. Across this body of work three limitations recur. First, all methods are unimodal, relying exclusively on Hi-C and discarding complementary evidence from sequencing coverage, sequence composition, and mappability. Second, all return coordinates without a calibrated reliability statement, making it impossible to distinguish well-supported predictions from noise-driven ones. Third, the tooling is fragmented, with preprocessing, inference, and scoring handled by separate programs with incompatible formats.

GenoFold answers all three gaps within one trained, end-to-end design [8]. It builds a contact graph whose nodes carry Hi-C contacts alongside per-bin sequencing coverage, $k$-mer composition, GC content, and mappability. The encoder pairs a four-layer GATv2 network with a six-layer Transformer, trained end-to-end with a self-supervised distance-matching objective. Reliability is a first-class output: a learned variance head with Monte Carlo dropout attaches a coordinate error radius to every position. Explanations come from the model itself through Integrated Gradients attributions and attention weights. The contributions are: (i) a trained multi-modal GATv2–Transformer reconstructor that matches established methods on dense data and outperforms them under sparse Hi-C; (ii) a head-to-head quantitative benchmark against 3DMax, HiC-GNN, FLAMINGO, and ShRec3D; (iii) validation against single-molecule FISH distances confirming physical accuracy; (iv) a sparse Hi-C experiment showing multi-modal features hold where the contact graph thins; and (v) per-bin uncertainty shown to correlate with actual reconstruction error.

TABLE I
NODE FEATURE COMPOSITION ($F = 140$).

| Feature group | Dim | Source |
|---|---|---|
| Coverage | 1 | RPKM read depth |
| $k$-mer sequence | 136 | RC-collapsed 4-mers (hg19) |
| Metadata | 2 | GC content, CRG mappability |
| Position | 1 | Normalized bin index |
| **Total** | **140** | |

## II. THE GENOFOLD FRAMEWORK

GenoFold is organized as six sequential stages with no manual intervention.

### A. Preprocessing

Raw Hi-C matrices are filtered to remove bins whose row-wise zero fraction exceeds 0.97, and are then normalized by iterative correction and eigenvector decomposition (ICE) [9] with convergence tolerance $10^{-5}$. This threshold is relaxed from the conventional 0.8 used in Hi-C analysis to accommodate centromeric and pericentromeric regions where sparsity is biologically expected. Bins filtered at this threshold are mainly unmappable repeat elements where no reconstruction method produces reliable coordinates; their exclusion is therefore consistent with standard practice across the methods compared against GenoFold. A distance-decay correction factor is then applied to isolate TAD structure from the distance baseline. Next-generation sequencing coverage is RPKM normalized, corrected for GC bias (linear regression) and mappability, and z-score transformed. GC content is calculated from the hg19 reference, and mappability is obtained from the ENCODE/CRG 50-mer alignability track.

### B. Graph Representation

The data are formatted as a contact graph $G = (V, E, X, E_{\text{attr}})$, where node features $X \in \mathbb{R}^{M\times 140}$ concatenate per-bin coverage (1D), reverse-complement-collapsed 4-mer frequencies (136D), normalized GC content and mappability (2D), and a positional encoding (1D) for each of the $M$ bins (Table I). Edge features encode the log-normalized contact weight and the log genomic distance. The graph is stored as a PyTorch Geometric `Data` object [10].

### C. Inference

Four Graph Attention Network v2 (GATv2) layers [11] update features by computing coefficients with which to attend to source, target, and edge features, ameliorating the static-attention limitation of GAT [12]. The first three layers employ 8 heads with concatenated outputs and residual connections, and the final layer averages over 4 heads to produce $d = 64$ features. Node embeddings are enriched with a six-layer pre-norm Transformer [13] that employs 8-head cross-attention (FFN dimension 512, GELU, dropout 0.1), sinusoidal position encoding, and global cross-bin interactions inaccessible to the GNN. A learned per-bin species embedding is included to aid future multi-species evaluation; in this work it is exercised on human cell lines only, and cross-species generalization is deferred to future work. Overall, the encoder contains 2,708,690 parameters.

### D. Reconstruction

A five-layer MLP ($64 \to 128 \to 256 \to 128 \to 3$) with batch normalization and GELU non-linearities decodes each node embedding to a set of 3D coordinates, constrained to lie within a specified range. Alternatively, stress-minimized multi-dimensional scaling (MDS) [14] with Procrustes analysis [15] can be used for coordinate reconstruction; reported results use the direct decoder output.

### E. Confidence and Uncertainty

A learned variance head predicts a per-bin, per-axis log-variance $v_{ik}$, converted to a standard deviation $\sigma_{ik} = \exp(v_{ik}/2)$. Monte Carlo dropout [16] runs $T$ stochastic passes, whose empirical standard deviation gives the epistemic term. The scalar per-bin radius

$$r_i = \Big(\sum\nolimits_k \sigma_{ik}^2\Big)^{1/2} \tag{1}$$

is the radius of a confidence ellipsoid around each predicted position. Structural domains are called by an insulation-score approach [17].

### F. Explainability

Integrated Gradients [18] are used to attribute the predictions to the four feature groups (coverage, $k$-mer sequence, metadata, position) by accumulating gradients along a 32-step path from the origin. In addition, the GAT and Transformer attention coefficients are used to compute inter-bin coupling as a function of bin-pair distance.

### G. Training Objective

Contact frequency $C_{ij}$ is mapped to a wish distance $D_{ij} = C_{ij}^{-0.25}$. The network minimizes a scale-invariant distance-matching loss over the set of observed pairs $P$:

$$\mathcal{L} = \frac{1}{|P|} \sum_{(i,j)\in P} \big(\hat{z}_{ij} - z^*_{ij}\big)^2, \tag{2}$$

where $\hat{z}_{ij}$ and $z^*_{ij}$ are the z-scored log reconstructed and wish distances, respectively. Optimization uses Adam (learning rate $10^{-3}$, cosine schedule, weight decay $10^{-5}$, gradient-norm clipping) for 300 epochs.

### H. Training Validation

Training convergence was monitored by held-out dSCC evaluated at every epoch on a 20% contact holdout set. The held-out dSCC stabilized within 150 epochs and plateaued for the remaining 150, with fewer than 0.002 dSCC variation across the final 50 epochs, indicating convergence prior to the end of training. No early stopping was applied, as the final checkpoint (epoch 300) was used for all reported evaluations. The training dSCC and held-out dSCC tracked within a few thousandths throughout, confirming that the model generalizes rather than simply memorizing observed pairs.

TABLE II
RECONSTRUCTION ACCURACY ON GM12878 CHR22 (701 BINS, 50 KB). BEST MEAN IN BOLD. ‡: FLAMINGO RUN BELOW ITS NATIVE RESOLUTION.

| Method | dSCC | TAD $F_1$ | Time (s) |
|---|---|---|---|
| ShRec3D | 0.887 | 0.45 | 1.4 |
| 3DMax | $\mathbf{0.900 \pm 0.000}$ | $0.59 \pm 0.01$ | 29.1 |
| HiC-GNN | $0.889 \pm 0.002$ | $\mathbf{0.60 \pm 0.01}$ | 100.6 |
| FLAMINGO‡ | $0.270 \pm 0.002$ | $0.50 \pm 0.00$ | — |
| GenoFold | $0.893 \pm 0.001$ | $0.57 \pm 0.01$ | 244.9 |

## III. EXPERIMENTAL SETUP

*Data.* We use the Rao et al. *in situ* Hi-C datasets [3] for GM12878, IMR90, and K562 from GEO accession GSE63525 at KR normalization. Hi-C coverage was used as a proxy for per-bin read depth; the hg19 reference genome was used to compute sequence $k$-mer statistics, GC content, and CRG mappability. All data are publicly available, and no synthetic features were used.

*Region and metric.* Unless stated otherwise, experiments use chromosome 22 at 50 kb resolution (701 valid bins after filtering). The primary metric is the distance Spearman correlation coefficient (dSCC), which is rank-invariant to the conversion exponent. TAD-boundary $F_1$ is also reported.

*Baselines.* GenoFold is compared against 3DMax [4], HiC-GNN [5], and ShRec3D [7], all re-implemented in a single framework consuming the identical contact submatrix. In addition, FLAMINGO [6] runs from its published R implementation. All are scored by the identical dSCC.

## IV. RESULTS

### A. Head-to-Head Benchmark

Table II summarizes results on GM12878 (chr22). Variability between runs was of the same order as differences between methods (about 0.01–0.02 dSCC), so means and standard deviations are reported. GenoFold achieves a dSCC of $0.893 \pm 0.001$, closely following HiC-GNN and matching 3DMax within a hundredth of a dSCC, with the four methods showing comparable qualitative behaviour. The weaker performance of FLAMINGO is explained by resolution: the method was designed for 1–5 kb Hi-C data, and results at 50 kb resolution are about an order of magnitude worse than at 5 kb. GenoFold recovers the cluster while providing additional information about uncertainty, modality contribution to prediction, and transferability of the embedding. The 244.9 s runtime reflects a single-invocation CPU implementation without optimization. The Transformer's $O(M^2)$ global attention and the Monte Carlo dropout passes account for the bulk of this cost; both are parallelizable and would reduce significantly on GPU hardware. Runtime optimization is the next focus.

### B. Validation Against Single-Molecule FISH Distances

While dSCC measures relative agreement with the training objective, we benchmarked the physical accuracy of the reconstructions by comparing predicted distances to super-resolution chromatin tracing distances [19]. Specifically, we used the publicly available median distance matrices for the IMR90 and K562 cell lines at chr21. GenoFold achieves a Spearman correlation of 0.986 (IMR90) and 0.981 (K562), comparable to 3DMax (0.987, 0.979) and better than raw Hi-C (0.973, 0.963).

### C. Reconstruction from Sparse Hi-C

At full depth, GenoFold and 3DMax agree with FISH almost equally; they separate once Hi-C is shallow. Downsampling each cell line's contacts and scoring against the full-depth FISH matrix, GenoFold's agreement is nearly flat: at 5% of contacts it reaches Spearman 0.978/0.969 (IMR90/K562), within a few thousandths of full depth. 3DMax falls to 0.819/0.831 and ShRec3D to 0.689/0.581, because methods that solve only from observed pairs have little left to fit once most pairs are gone. GenoFold's node features (coverage, sequence, GC content, mappability) do not thin with sequencing depth, and its learned prior supplies structure where contacts are absent. This is the regime where a trained multi-modal reconstructor earns its cost.

### D. Additional Results

A modality ablation confirms that at full depth all configurations fall within roughly 0.01 dSCC: the dense contact graph dominates and node modalities are near-redundant. The multi-modal advantage grows monotonically as contacts are removed (from $+0.002$ at half depth to $+0.005$ at 5%), consistent with the FISH result. Holding out 20% of pairs, the held-out dSCC (0.890) tracks the training dSCC to within a few thousandths, confirming that the model is not memorizing pairs. Retrained on IMR90 and K562, GenoFold reaches dSCC 0.795 and 0.754; a GM12878-trained model transfers to these lines without retraining at 0.630 and 0.653, well above chance. Combined uncertainty correlates with per-bin reconstruction error at Spearman 0.28.

## V. DISCUSSION

The results establish that a single trained architecture can carry multi-modal genomic inputs from raw matrices to an annotated, uncertainty-tagged 3D structure at accuracy competitive with specialized methods. On dense Hi-C data, the field's methods cluster within run-to-run noise. On shallow Hi-C, GenoFold recovers physically measured structure far more faithfully than distance optimizers, because its node features and learned prior do not depend on a dense contact graph. Three further capabilities separate it from the baselines: it generalizes to held-out contacts and transfers across cell lines without re-solving; its per-bin uncertainty correlates with actual error; and its feature-grouped node representation enables modality-level attribution from a trained model, something none of the baselines can provide. These properties all draw on one shared representation: the confidence head reads the same embeddings the decoder turns into coordinates, and attribution is meaningful only because features were kept grouped by biological source.

*Threats to validity.* The self-supervised objective trains against Hi-C-derived distances rather than resolved structures; Section IV-B confronts this with FISH validation, though only over one 2 Mb region. FLAMINGO is run below its native resolution in the main table, and most of that gap is recovered at 5 kb. The evaluated cell lines are all human, so the species-embedding layer is exercised on a single species. The 0.97 zero-fraction filtering threshold retains biologically sparse regions including centromeric sequences where mappability is low; however, reconstruction accuracy in these regions may still be limited by contact sparsity regardless of filtering choice.

## VI. Conclusion

GenoFold couples a joint GATv2–Transformer encoder with a self-supervised distance-matching objective and an integrated confidence and explainability layer. On real *in situ* Hi-C data it reconstructs chromatin structure competitively with 3DMax, HiC-GNN, and ShRec3D, an agreement that single-molecule FISH distances confirm is physically accurate (Spearman 0.986 and 0.981). Alongside that score it uniquely adds per-bin uncertainty correlating with error, multi-modal features that grow more useful as Hi-C is sparsified, cross-cell-line transfer without retraining, and whole-chromosome scaling on commodity CPU. Runtime optimization through GPU deployment and extension of FISH validation beyond a single genomic region are the vital next steps. Code, data loaders, and trained checkpoints will be released upon acceptance.

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